

Spring 2026 Seminar on Kac–Moody Lie Algebras

Organized by students

Saturdays, 19:00–21:30
Room 233B, Taizhou Building

Date	Speaker	Content
Mar 21	Kanghe	Definitions and first properties – [K, §1.1–1.4]
Mar 28	Huimin	Triangular decomposition, root spaces, Chevalley involution – [K, §1.5–1.8]
Apr 04	Binhe	Invariant bilinear form: construction and properties – [K, §2.1–2.4]
Apr 11	Yiliang	Casimir operator; integrable modules – [K, §2.5–2.8]
Apr 18	TBA	More on invariant forms; Heisenberg algebra – [K, §2.9–2.11]
Apr 25	TBA	Weyl group: definitions, local nilpotency, fundamental reflections – [K, §3.1–3.4]
May 02	TBA	Weyl group: exponentials, integrable modules, W as Coxeter group – [K, §3.5–3.8]
May 09	TBA	Reduced expressions, exchange condition, Tits cone – [K, §3.9–3.12]
May 16	TBA	Coxeter structure, fundamental chamber, classification of GCMs I – [K, §3.13–3.15]
May 23	TBA	Classification of GCMs II: finite, affine, indefinite type – [K, §4.1–4.5]
May 30	TBA	Dynkin diagrams, finite and affine tables – [K, §4.6–4.9]
Jun 06	TBA	Further properties of GCMs; real and imaginary roots (I) – [K, §4.10–4.11 & §5.1–5.2]
Jun 13	TBA	Real and imaginary roots (II): support, description of Δ^{im} – [K, §5.3–5.4]

Main references

- [K] Kac, V. G., *Infinite-Dimensional Lie Algebras*, 3rd ed., Cambridge University Press, 1990.
- [C] Carter, R. W., *Lie Algebras of Finite and Affine Type*, Cambridge University Press, 2005.